Mid 1 Important Questions (Set 1 and Set 2)

Part A: Short Answer Questions

10 Questions × 2 Marks = 20 Marks (Combined)

Q.NO	SHORT ANSWER QUESTION (2 MARKS)	SOURCE
1.	Draw a neat labeled diagram of a coaxial cable, clearly showing its four main structural layers	Set 1
2.	What is WDM?	Set 1
3.	With an example explain error detection using Hamming code	Set 1
4.	What are the design issues of data link layer?	Set 1
5.	What is flooding?	Set 1
6.	Describe unguided transmission media with two examples	Set 2
7.	Write any two key differences between FDM and TDM	Set 2
8.	A data word **`1011001`** is transmitted using even parity. Find the parity bit that should be added to the data. At the receiver, the received data is **`1010001`**. Determine whether a single-bit error has occurred	Set 2
9.	What is the purpose of framing in data link layer?	Set 2
10.	What is virtual circuit?	Set 2

Part B: Long Answer Questions

3 × 10 = 30 Marks (Per Set)

Q.NO	LONG ANSWER QUESTION (DUPLICATE / OVERLAPPING QUESTIONS COMBINED)	SOURCE
1.	Discuss in detail the layers of TCP/IP with a neat diagram	Set 1
2.	What is network topology? Explain any three network topologies with neat diagrams along with their limitations. **Combined with Set 2:** Describe the characteristics and limitations of network topologies—star, bus, and ring.	Set 1 & 2
3.	Compare OSI and TCP/IP reference model	Set 1 (OR)

Q.NO	LONG ANSWER QUESTION (DUPLICATE / OVERLAPPING QUESTIONS COMBINED)	SOURCE
4.	With neat sketches explain frequency division multiplexing (FDM) and time division multiplexing (TDM)	Set 1 (OR)
5.	Explain in detail any three functionalities of all the layers in OSI model	Set 2
6.	Discuss in detail any two copper media with neat diagrams	Set 2
7.	Differentiate between LAN and WAN based on geographical coverage, ownership, transmission speed, error rate	Set 2 (OR)
8.	Discuss sliding window protocol using Go-Back-N	Set 1
9.	Discuss in detail the sliding window protocol using Selective Repeat	Set 2
10.	Explain the working of CSMA/CD	Set 1
11.	What is the purpose of CSMA with collision prevention?	Set 2 (OR)
12.	Explain how Slotted ALOHA solves the problem of channel allocation **Combined with Set 2:** Explain in detail the operation of Pure ALOHA and Slotted ALOHA	Set 1 & 2
13.	Discuss in detail the working of Token Bus.	Set 1 (OR)
14.	What is CRC? Explain CRC algorithm with a example	Set 2
15.	Explain about Distance Vector Routing with an example	Set 1
16.	With a suitable explain link state routing algorithm	Set 2
17.	Explain the Dijkstras shortest path algorithm with a suitable example	Set 2 (OR)